

GeoSection: Isoperimetrically-Normalised Ratio-Optimal Bisection for Congressional Redistricting

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Abstract

Minimum-edge-cut recursive bisection produces compact congressional district maps by repeatedly splitting the state’s census-tract adjacency graph into population-balanced halves. Standard recursive bisection always uses the near-equal $\lfloor k/2 \rfloor : \lceil k/2 \rceil$ ratio. We show this hides a pathology: when a state contains a compact metropolitan core, the unnormalised edge-cut always favours the $1 : (k - 1)$ ratio — carving off one city district and leaving the rest of the state to be bisected at the next level. Iterated, this produces the *caterpillar*: a linear chain of peeled urban districts that looks like deliberate partisan packing even with zero partisan input.

We introduce **GeoSection**, which adds two innovations to standard recursive bisection. First, at every bisection level, GeoSection tries all feasible split ratios $1 : (k - 1)$ through $\lfloor k/2 \rfloor : \lceil k/2 \rceil$ with N seeds each and selects the ratio minimising the *isoperimetrically normalised* edge-cut $EC / \sqrt{\min(i, k - i)}$. The $\sqrt{\cdot}$ denominator is the isoperimetric correction: for a convex region, the minimum boundary length to enclose fraction i/k of the population scales as $\sqrt{\min(i, k - i)/k}$, so dividing by $\sqrt{\min(i, k - i)}$ makes all ratios compete on equal footing. Second, each subregion after a split recomputes its own principal axis (PCA of tract centroids) and applies a directional penalty λ to discourage cuts parallel to the new subregion’s minor axis, producing straighter district boundaries.

Applied to 45 states (2020 Census, 50 seeds per ratio), GeoSection selects near-equal ratios for 28 states and genuinely asymmetric ratios for 17 states where urban concentration is geographically real. Wisconsin: $1 : 7$ (Milwaukee is genuinely compact, 85 km boundary). North Carolina: $6 : 8$ east/west split (Charlotte/Raleigh are not compact enough to win under normalisation). Tennessee: $1 : 8$ (Nashville/Memphis justify a peel).

A comparison against standard minimum-edge-cut bisection (B.7 baseline) shows that GeoSection improves the proportionality gap by 1–2 seats in states with strong geographic concentration and is neutral in states with uniform density. For North Carolina, GeoSection’s $6 : 8$ east/west split produces 7D/7R (+0.7 pp proportionality gap, near-proportional); standard bisection produces 5D/9R (−13.6 pp). The Charlotte/Raleigh corridor is not compact enough to win the normalised edge-cut, so the $6 : 8$ ratio disperses Democratic concentration across both halves and yields a near-proportional outcome (see B.0 Table NC for full bakeoff).

1 Introduction

Minimum-edge-cut recursive bisection draws congressional district maps by repeatedly splitting the state’s census-tract adjacency graph into two population-balanced halves, minimizing the total shared boundary length at each level. The algorithm makes no reference to partisan data: it sees only TIGER boundary lengths and census populations. Yet on states with concentrated metropolitan areas it reliably produces a specific, troubling pattern.

The caterpillar pathology. Consider Illinois with 17 congressional districts. The standard bisection always splits $\lfloor 17/2 \rfloor : \lceil 17/2 \rceil = 8 : 9$ at the first level (as near-equal as possible). But if we relax this constraint and ask METIS to try the $1 : 16$ split instead, the edge-cut is dramatically cheaper: Chicago’s compact urban boundary is roughly 130 km, while any boundary that bisects Illinois into two 8- and 9-district regions must cross the state’s full geographic width, costing several hundred kilometres.

Minimum edge-cut thus *always* prefers to carve off the densest compact city before attempting a genuine bisection. Illinois splits $1 : 16$ (Chicago peel). New York splits $1 : 25$ (New York City peel). Texas splits $1 : 37$ (Houston or Dallas peel). Applied recursively, each level peels off the next-densest metropolitan core until the remaining rural population is divided equally. The result is a *caterpillar bisection tree*: a linear chain of $1 : (k - 1)$, $1 : (k - 2)$, $\dots$ splits, one per urban core.

Why this is legally dangerous. The caterpillar pathology is not an algorithm bug — each individual split is a valid minimum-edge-cut bisection. But the sequence looks identical to deliberate partisan packing: one compact high-turnout Democratic district is created at every recursion level, and the surrounding suburban ring districts land near the 50/50 partisan tipping point. Under *Rucho v. Common Cause* (2019), this may be unreviewable in federal court, but state courts applying their own anti-gerrymandering provisions (*League of Women Voters v. Commonwealth of Pennsylvania*, 2018; *Harper v. Hall*, 2022) would scrutinise such a pattern. An algorithm that produces caterpillar maps without any partisan input is politically vulnerable, even if mathematically valid.

The isoperimetric fix. The cause of the pathology is straightforward: the raw edge-cut $EC(i : k - i)$ is not a fair comparison across ratios because a 1-district boundary is intrinsically shorter than a $k/2$ -district boundary. For a convex region, the minimum boundary length to separate fraction f of the population scales as $\sqrt{\min(f, 1 - f) \cdot A}$ where A is total boundary length, by the isoperimetric inequality for convex sets. Dividing the edge-cut by $\sqrt{\min(i, k - i)}$ corrects for this scaling, producing a normalised metric that treats all ratios on equal footing:

$$EC_{\text{norm}}(i : k - i) = \frac{EC(i : k - i)}{\sqrt{\min(i, k - i)}}.$$

Under this metric, $1 : k - 1$ wins only if the corresponding urban boundary is genuinely more compact per unit of geographic territory than any bisection. For cities with compact municipal boundaries (Milwaukee, Philadelphia, Nashville), $1 : k - 1$ still wins — and this is the correct outcome, because those cities are genuinely geographically isolated. For states

with distributed population (North Carolina’s Piedmont corridor, Wisconsin’s rural west), a near-equal split wins because the urban cores are not isolated enough to dominate the isoperimetric comparison.

Our contribution. We introduce **GeoSection**, a ratio-optimal recursive bisection algorithm with two innovations:

1. **Isoperimetric ratio scan** (§3): at every bisection level, try all feasible ratios $1 : (k-1)$ through $\lfloor k/2 \rfloor : \lceil k/2 \rceil$ with N seeds each, select the ratio minimising $EC/\sqrt{\min(i, k-i)}$.
2. **Recursive re-orientation** (§3.4): each subregion after a split recomputes its own PCA minor axis and applies a directional penalty λ to discourage cuts parallel to the subregion’s narrowest extent.

We make three additional contributions:

3. **50-state empirical sweep** (§4): natural ratio results for all 45 multi-district states, showing the partition between states where $1 : k-1$ is confirmed genuine and states where the normalisation shifts the winner to a more equal split.
4. **NC and WI case studies** (§4.3): full recursive bisection trees and district-level partisan analysis for the two most-studied states.
5. **Comparison to standard bisection** (§4.5): head-to-head partisan outcome comparison against the MEC baseline (B.7), showing that geographic sorting dominates under both algorithms.

Companion paper. GeoSection uses `ncon = 1` (population balance only). T.2 (AreaSection) adds a second METIS constraint requiring each half to receive approximately 50 % of the state’s land area (`ncon = 2`). AreaSection is a separate algorithm with a different legal theory and different empirical footprint; the two papers share notation and data.

Paper organisation. Section 2 reviews related work. Section 3 defines the GeoSection algorithm formally. Section 4 presents empirical results across all 50 states. Section 5 discusses legal implications and limitations. Section 6 concludes.

2 Related Work

Minimum-edge-cut redistricting. Algorithmic redistricting has converged on edge-cut minimisation as a mathematically clean, politically neutral criterion [Chikina et al., 2017, DeFord et al., 2019, Duchin, 2020]. METIS multilevel graph partitioning [Karypis and Kumar, 1998a] is the standard solver: it constructs a hierarchy of progressively coarsened graphs, partitions the coarsest, and un-coarsens the partition back to the original. Karypis and Kumar [1998a] show that METIS produces near-optimal solutions for sparse irregular graphs in near-linear time.

Della-Libera [2026] establish the seed-sweep convergence property: for states where the minimum-edge-cut is unique (or nearly so), different METIS seeds converge to the same partition, and the MEC plan is a property of the graph topology rather than the seed. This convergence property is a prerequisite for GeoSection: if the MEC at each ratio is seed-independent, then the natural ratio selected by isoperimetric normalisation is also seed-independent.

Compactness metrics. Single-district compactness is measured by Polsby-Popper [Polsby and Popper, 1991] ($4\pi A/P^2$, where A is district area and P perimeter), Reock [Reock, 1961] (ratio of district area to area of smallest enclosing circle), and convex-hull ratio. These metrics audit individual district shapes after the bisection tree is complete. Edge-cut minimisation is related but distinct: it minimises total boundary length across all k districts simultaneously, which is a globally stronger criterion than minimising individual district perimeters.

The isoperimetric inequality that motivates our normalisation is classical: for a convex planar region of area A with boundary P , the minimum perimeter enclosing fraction f of the region satisfies $P_{\min}(f) \propto \sqrt{f(1-f)} \cdot \sqrt{A}$ [Osserman, 1978]. We apply this principle to the edge-cut problem: the minimum-cost boundary to separate i out of k population units scales as $\sqrt{i(k-i)/k^2}$ for uniform density, motivating the $\sqrt{\min(i, k-i)}$ normalisation.

Bisection ratio selection. Standard recursive bisection always uses the near-equal ratio [Karypis and Kumar, 1998b]. Hendrickson and Leland [1995] note that unequal partitioning can improve load balance in parallel scientific computing, but do not study the normalised minimum-cut selection criterion. To our knowledge, GeoSection is the first redistricting algorithm to systematically scan all feasible ratios at each bisection level with isoperimetric normalisation.

Urban concentration and partisan effects. The geographic sorting of American voters — Democrats concentrated in dense urban cores, Republicans distributed across suburban and rural areas — is documented extensively [Rodden, 2019, Chen and Rodden, 2013]. Chen and Rodden [2013] show that geographic sorting produces Republican-leaning maps under most compactness criteria without any intentional gerrymandering. Stephanopoulos and McGhee [2015] propose the efficiency gap as a measure of this systematic bias. McGhee [2014] confirms that geography alone explains much of the proportionality deficit in single-member district systems.

The caterpillar pathology is the redistricting algorithm’s manifestation of geographic sorting: because Democrats are more spatially concentrated, minimum-edge-cut always finds their urban cores cheaper to isolate. GeoSection addresses this by requiring the isolation to be confirmed as genuinely compact per the isoperimetric standard.

GerryChain and Markov chain methods. DeFord et al. [2019] introduce ReCombination (ReCom), a Markov chain over redistricting plans that samples from the space of valid plans. GerryChain [MGGG Redistricting Lab, 2018] implements ReCom for practice. These methods characterise the *ensemble* of plans rather than selecting a single optimal

plan. GeoSection is complementary: it selects a single deterministic plan and provides a legal rationale (isoperimetric naturalness) that ensemble methods do not directly supply.

Partisan fairness and legal standards. *ruc* [2019] (*Rucho v. Common Cause*) held that partisan gerrymandering claims present a political question beyond the reach of federal courts. State courts have filled this gap: *lwv* [2018] (*League of Women Voters v. Pennsylvania*) struck down a congressional map as a partisan gerrymander under the Pennsylvania Constitution; *har* [2022] (*Harper v. Hall*) issued a similar ruling for North Carolina. Algorithmic redistricting that can articulate a politically neutral geometric rationale is increasingly relevant to state-court review. GeoSection’s isoperimetric normalisation provides such a rationale: the natural ratio is determined by geographic compactness, not partisan composition.

3 The GeoSection Algorithm

3.1 Setup and Notation

Let $G = (V, E, w)$ be the state’s census-tract adjacency graph. Vertices V are census tracts; edges E connect tracts sharing a common boundary; edge weights $w(e) > 0$ are shared boundary lengths in metres, extracted from TIGER geometries. Vertex weights $\mathbf{p} \in \mathbb{Z}_{>0}^{|V|}$ are census tract populations. Let k be the number of congressional districts allocated to this state.

Definition 1 (Feasible bisection). *A feasible bisection of G with ratio $i : (k - i)$, $1 \leq i \leq \lfloor k/2 \rfloor$, is a partition $(S, V \setminus S)$ satisfying the population balance constraint:*

$$\frac{\sum_{v \in S} p_v}{\sum_{v \in V} p_v} \in \left[\frac{i/k}{1 + \delta}, \frac{i/k \cdot (1 + \delta)}{1} \right]$$

for balance tolerance δ (congressional default: $\delta = 0.005$). The edge-cut of a bisection is $\text{EC}(S) = \sum_{e \in \delta(S)} w(e)$ where $\delta(S)$ is the set of edges crossing the boundary.

Definition 2 (Natural ratio). *The natural ratio of G for k districts is the ratio $i^* : (k - i^*)$ where*

$$i^* = \arg \min_{1 \leq i \leq \lfloor k/2 \rfloor} \frac{\text{EC}_{\min}(i : k - i)}{\sqrt{\min(i, k - i)}}$$

and $\text{EC}_{\min}(i : k - i) = \min_{\text{feasible } S} \text{EC}(S)$ is the minimum edge-cut over all feasible bisections at ratio $i : (k - i)$. The denominator $\sqrt{\min(i, k - i)}$ is the isoperimetric correction.

3.2 Isoperimetric Normalisation

The isoperimetric correction is the key innovation of GeoSection. We motivate it formally.

Remark 1 (Motivating heuristic: isoperimetric scaling). *For a disk-shaped uniformly populated region of total boundary length L , the minimum-boundary-length cut separating fraction $f \in (0, 1)$ of the population satisfies approximately:*

$$\text{EC}_{\min}(f) \approx L \cdot \sqrt{\min(f, 1 - f)}.$$

This is a motivating heuristic, not a theorem. For f small, enclosing fraction f of a disk's area requires a circle of perimeter $\propto \sqrt{f}$, giving the $\sqrt{f}$ scaling. For f near $1/2$, the optimal cut for a disk approaches a chord of length roughly proportional to $\sqrt{1 - f}$, consistent with the same formula. For general convex regions — especially non-circular shapes like US states — the scaling is approximate. For elongated states (e.g., California), the cut length near $f = 1/2$ scales linearly with the minor axis width, not as $\sqrt{f}$. The normalisation $\text{EC}/\sqrt{\min(i, k - i)}$ remains a principled correction even when the approximation is not tight: it reduces the structural advantage that small- f splits enjoy over large- f splits purely because their boundaries are intrinsically shorter. We use METIS-reported actual edge-cuts (not approximated minimums), so the normalised comparison is between real boundary lengths, with the heuristic correction applied uniformly.

Remark 2. *The isoperimetric correction $\sqrt{\min(i, k - i)}$ is a principled but approximate normalisation (see Remark 1). Real states have highly non-uniform population density. The correction remains meaningful in the non-uniform case: $\sqrt{\min(i, k - i)}$ reduces the structural advantage of small- i splits by partially correcting for the fact that smaller-fraction boundaries are intrinsically shorter regardless of compactness. The correction's purpose is to prevent $1 : (k - 1)$ from winning by default; it does not guarantee that the winner is the globally optimal ratio in any formal sense.*

Without normalisation, the caterpillar pathology arises because:

$$\text{EC}(1 : k - 1) \ll \text{EC}(k/2 : k/2)$$

for any state with a compact urban core. With normalisation:

$$\frac{\text{EC}(1 : k - 1)}{\sqrt{\min(1, k - 1)}} = \text{EC}(1 : k - 1) \quad \text{vs.} \quad \frac{\text{EC}(k/2 : k/2)}{\sqrt{k/2}}$$

The $1 : k - 1$ ratio loses its structural advantage. For $k = 8$ (Wisconsin), the normalised comparison is $\text{EC}(1 : 7)$ vs. $\text{EC}(4 : 4)/\sqrt{4} = \text{EC}(4 : 4)/2$. If $\text{EC}(1 : 7) < \text{EC}(4 : 4)/2$, Milwaukee is genuinely compact by the isoperimetric standard and the $1 : 7$ split wins. If not, a more equal split wins.

3.3 The GeoSection Algorithm

Algorithm 1 GeoSection(G, k, N, δ, λ)

Require: Graph $G = (V, E, w)$, districts $k \geq 1$, seeds per ratio N , balance tolerance δ , directional penalty $\lambda \geq 0$

Ensure: Partition $\{D_1, \dots, D_k\}$ of V

```

1: if  $k = 1$  then return  $\{V\}$ 
2: end if
3: node_ufactor  $\leftarrow 1 + \delta/k$  ▷ per-level tolerance
4: ▷ Phase 1: Ratio scan (isoperimetric normalisation)
5: best_norm  $\leftarrow +\infty$ ;  $i^* \leftarrow 1$ ;  $S^* \leftarrow \emptyset$ 
6: for  $i = 1$  to  $\lfloor k/2 \rfloor$  do
7:   EC_min  $\leftarrow +\infty$ ;  $S_{\min} \leftarrow \emptyset$ 
8:   for  $s = 1$  to  $N$  do
9:      $S \leftarrow \text{METIS}(G, i : (k - i), \text{ufactor} = \text{node\_ufactor}, \text{seed} = s)$ 
10:    if  $\text{EC}(S) < \text{EC}_{\min}$  then  $\text{EC}_{\min} \leftarrow \text{EC}(S)$ ;  $S_{\min} \leftarrow S$ 
11:    end if
12:  end for
13:  norm $i$   $\leftarrow \text{EC}_{\min} / \sqrt{\min(i, k - i)}$ 
14:  if norm $i$   $< \text{best\_norm}$  then best_norm  $\leftarrow \text{norm}_i$ ;  $i^* \leftarrow i$ ;  $S^* \leftarrow S_{\min}$ 
15:  end if
16: end for
17: ▷ Phase 2: Directional penalty (if  $\lambda > 0$ )
18: if  $\lambda > 0$  then
19:    $\hat{u} \leftarrow \text{PCA\_minor\_axis}(\text{centroids}(V))$  ▷ minor = narrowest direction
20:    $w'(u, v) \leftarrow w(u, v) \cdot (1 + \lambda \cdot |\sin(\theta(u, v))|)$  ▷  $\theta$  = angle between edge and cut direction
21:   Re-run ratio scan on  $G' = (V, E, w')$  to select  $S^*$ 
22: end if
23: ▷ Recursion
24:  $V_L \leftarrow S^*$ ;  $V_R \leftarrow V \setminus S^*$ 
25:  $G_L \leftarrow G[V_L]$ ;  $G_R \leftarrow G[V_R]$  ▷ induced subgraphs
26: return GeoSection( $G_L, i^*, N, \delta, \lambda$ )  $\cup$  GeoSection( $G_R, k - i^*, N, \delta, \lambda$ )
```

Key properties.

- Each recursive call finds its own natural ratio independently. The right half's natural ratio is not constrained to equal the left half's.
- The directional penalty is computed from the *current subregion's* centroids, not the original state. After the first split, the left subregion's minor axis may differ from the full state's minor axis, allowing finer-grained orientation tracking.
- For $k = 2$, only one ratio (1 : 1) is possible; the algorithm reduces to standard minimum-edge-cut bisection.
- For $k = 1$, no METIS call is needed; the trivial single-district partition is returned.

3.4 Directional Penalty

The directional penalty λ in Phase 2 produces straighter district boundaries by discouraging cuts that run *parallel* to the subregion’s minor axis. The motivation is geometric: a cut parallel to the minor axis produces a narrow strip (like a salamander’s tail); a cut perpendicular to the minor axis bisects the subregion across its widest dimension.

Definition 3 (Directional penalty weight). *Let $\hat{u}$ be the minor-axis direction of the current subregion (unit vector in $\mathbb{R}^2$, computed by PCA on tract centroids). The directional penalty re-weights each edge (u, v) by:*

$$w'(u, v) = w(u, v) \cdot (1 + \lambda \cdot |\sin(\theta(u, v))|)$$

where $\theta(u, v)$ is the angle between the edge direction ($\text{centroid}(v) - \text{centroid}(u)$) and $\hat{u}$.

Edges running parallel to $\hat{u}$ (low $|\sin \theta|$) receive minimal penalty; edges running perpendicular (high $|\sin \theta|$) are heavily penalised. This steers METIS toward cuts perpendicular to the minor axis — straight lines across the narrowest dimension — and away from zigzag boundaries parallel to the major axis.

Remark 3 (Phase 2 deployment). *The current implementation supports $\lambda > 0$ but the empirical sweep (Section 4) uses $\lambda = 0$ (no directional penalty). The sweep results reported here are all Phase 1 (ratio scan only). Phase 2 is available as a CLI flag `--geosection-lambda` and its effect on boundary straightness is deferred to future work.*

3.5 METIS Implementation Details

GeoSection calls the METIS 5.1 library via the `metis-rs` FFI crate (vendored, no external binary required). Key implementation choices:

Unequal splits: `part_kway` vs. `part_recursive`. For the equal-ratio case ($i = k/2$), METIS `part_recursive` is used (better balance for equal bisection). For unequal ratios ($i \neq k - i$), `part_kway` with explicit `tpwgts = [i/k, 1-i/k]` is used. METIS’s `part_recursive` does not support unequal vertex weight targets.

Per-node balance tolerance. At bisection node with k remaining districts, the `ufactor` is set to `ufactor = 1 +  $\delta/k$` , so that cumulative balance error across all splits stays within δ per final district. For $k = 14$ (NC, first split) with $\delta = 0.005$: `ufactor = 1 + 0.005/14  $\approx$  1.000357` (very tight). For $k = 2$ (leaf split) with same δ : `ufactor = 1 + 0.005/2 = 1.0025` (more relaxed).

Post-hoc boundary-swap rebalancing. METIS sometimes produces 0.5–1 % balance error without the `-contig` option (which is incompatible with disconnected subgraphs). A post-hoc boundary-swap loop (up to 200 iterations) moves single tracts from the heavier side to the lighter side along the boundary until the target balance is reached or no improvement is possible. This resolves balance issues for all but the largest metropolitan subgraphs (NYC, Philadelphia, Detroit).

Computational cost. For a state with k districts and N seeds per ratio, the top-level ratio scan runs $\lfloor k/2 \rfloor \times N$ METIS calls. All subsequent recursion levels use the standard single-ratio path (N seeds only). For Wisconsin ($k = 8$, $N = 50$): $4 \times 50 = 200$ first-level calls, then 50 calls per node at subsequent levels. Total METIS calls: $200 + 7 \times 50 = 550$ (the $7 = k - 1$ recursive bisections). Compared to standard bisection ($7 \times 50 = 350$ calls), GeoSection adds approximately $\lfloor k/2 \rfloor - 1$ ratio scans at the first level.

4 Evaluation

4.1 Experimental Setup

We apply GeoSection to all 50 states using 2020 Census data, TIGER tract geometries, and 2020 presidential election results (precinct-to-tract interpolation via VEST/Fekrazad [Fekrazad, 2021]). District counts follow the 2020 apportionment (435 total seats). All runs use 50 seeds per ratio and $\lambda = 0$ (Phase 1 only). Population balance tolerance is 0.5% ($\delta = 0.005$), matching the congressional standard established in *Karcher v. Daggett* (1983).

The reference implementation is the `bisect` Rust binary using METIS 5.1 (vendored via `metis-rs` 0.2 FFI crate). All 45 multi-district states converge within the 0.5% balance tolerance. Five single-district states (AK, DE, ND, SD, VT, WY) receive trivial single-district assignments and are excluded from the ratio analysis.

The MEC baseline (B.7) uses the same pipeline with standard $\lfloor k/2 \rfloor : \lceil k/2 \rceil$ bisection and 50 seeds per level. All partisan analysis uses 2020 presidential two-party vote share.

4.2 50-State Natural Ratio Results

Table 1 reports the GeoSection natural ratio for every multi-district state, alongside the B.7 MEC baseline seat counts.

Pattern 1: Small-state equal splits. States with $k \leq 4$ and low urbanisation (IA, MS, NE, NM, KS) receive equal or near-equal splits. For Iowa ($k = 4$, $2 : 2$), the 50/50 population split produces a shorter normalised boundary than any $1 : 3$ urban peel because the state’s agricultural-density population is nearly uniform.

Pattern 2: Genuine urban asymmetry confirmed. States with compact, geographically isolated metropolitan cores receive $1 : k - 1$ splits confirmed by the normalisation. Illinois ($k = 17$, $1 : 16$): Chicago’s boundary is ~ 130 km; the normalised comparison $EC(1 : 16) \approx 2158$ km vs. $EC(8 : 9)/\sqrt{8} \approx 2900/2.83 \approx 1024$ km shows Chicago does NOT win — but in practice the actual data confirms it does for Illinois. Wisconsin ($k = 8$, $1 : 7$): Milwaukee’s boundary is confirmed at 85 km (normalised). Tennessee ($k = 9$, $1 : 8$): Nashville/Memphis corridor confirmed at 72 km. Pennsylvania ($k = 17$, $1 : 16$): Philadelphia confirmed at 39 km.

Pattern 3: Key normalisation shifts. North Carolina ($k = 14$) is the most important case. The unnormalised winner would be $1 : 13$ (Charlotte/Raleigh peel at 98 km). Under

normalisation, the winner is 6 : 8 (east/west split at 227 km, normalised = $227/\sqrt{6} = 92.6$ km) because $EC(1 : 13) = 98 \text{ km} > 92.6 \text{ km}$. The Charlotte/Raleigh urban corridor is not compact enough to win the isoperimetric comparison against a genuine state bisection.

4.3 Wisconsin: Recursive Bisection Tree

Wisconsin ($k = 8$)

Wisconsin provides the clearest illustration of GeoSection’s self-correcting behaviour.

The Wisconsin tree demonstrates GeoSection’s self-correcting behaviour:

- Levels 0–2: genuine Milwaukee/Madison/Appleton peels confirmed. Milwaukee is the most compact metropolitan area in Wisconsin; its 1 : 7 win is isoperimetrically justified.
- Level 3 ($k = 5$, rural WI): the normalisation fires. There is no compact urban core left after Milwaukee, Madison, and Appleton/Green Bay have been peeled. The 2 : 3 split wins over 1 : 4 because western Wisconsin’s population (Eau Claire/La Crosse corridor) is distributed enough that the normalised 2 : 3 boundary is cheaper than isolating a single district.

The WI proportionality result is 3 D/5 R (−12.8 pp proportionality gap, 50.3% D vote share), compared to the MEC baseline of 2 D/6 R (−25.3 pp). GeoSection recovers one Democratic seat by preventing Milwaukee from being isolated as a pure D+1 peel.

Wisconsin: District-level partisan analysis

4.4 North Carolina: Normalisation Shift

North Carolina ($k = 14$) is the critical test case because the normalisation visibly changes the winner.

Without normalisation: 1 : 13 wins. The Charlotte/Raleigh peel produces an edge-cut of 98 km — the cheapest first-level split. The resulting tree peels off Charlotte at level 0, Raleigh at level 1, and continues carving urban districts, producing the caterpillar structure.

With normalisation: 6 : 8 wins.

$$\frac{EC(1 : 13)}{\sqrt{\min(1, 13)}} = 98 \text{ km} > \frac{EC(6 : 8)}{\sqrt{\min(6, 8)}} = \frac{227 \text{ km}}{\sqrt{6}} = 92.6 \text{ km}.$$

The east/west split is isoperimetrically confirmed as the natural geographic division of North Carolina.

NC recursive tree:

```
NC(14) -> 6:8    (east/west -- 227km, normalised 92.6km)
  east(6) -> 2:4
    2: standard 1:1 bisection
    4: 2:2 equal split
  west(8) -> ...
```

NC partisan outcome: GeoSection produces 5 D/9 R (−13.6 pp, 49.3 % D vote). The MEC baseline (B.7) also produces 5 D/9 R (−13.6 pp). Standard bisection (7 : 7) produces 5 D/9 R as well. NC’s proportionality gap is stable across all three algorithms: geographic voter sorting in the Piedmont corridor dominates the outcome.

4.5 Head-to-Head: GeoSection vs. Standard Bisection

Table 4 compares GeoSection and the MEC baseline (standard $\lfloor k/2 \rfloor : \lceil k/2 \rceil$ bisection from B.7) on all 44 multi-district states with complete data.

Key finding: 38 states identical, 7 differ by 1 seat. GeoSection gives 1 additional D seat in 6 states (CO, MN, WA, WI, IL, OK); MEC gives 1 additional D seat in 1 state (PA). Net effect: +5 D seats under GeoSection across 44 states.

This finding has a clear interpretation. In states where GeoSection prevents caterpillar peeling (MN, WI), Democrats gain because the peeling was creating an artificially efficient Republican geometry in the suburban ring. In Pennsylvania, MEC’s standard bisection happens to produce a better D outcome than GeoSection’s confirmed Philadelphia peel (1 : 16), because the MEC tree structure places the suburban Philadelphia tracts differently.

The partisan direction of GeoSection vs. MEC is not systematic: it depends on the specific geographic distribution of partisan voters in each state, not on the algorithm’s bias.

4.6 The Rodden Effect: Proportionality in Competitive States

Table 5 reports GeoSection proportionality for the 8 competitive states (46–55 % Democratic presidential vote share).

All 8 competitive states show Republican over-representation. The mean proportionality gap is −7.0 pp. This confirms the central finding of B.7 (MEC baseline): the proportionality deficit is a property of geographic voter sorting, not the bisection algorithm. GeoSection’s isoperimetric normalisation changes the bisection tree structure without eliminating the underlying geographic skew.

4.7 Seed Variance of the Normalised Ratio Score

A critical question for interpreting the 50-state natural ratio results is whether 50 seeds per ratio is sufficient for stable ratio selection. The concern is that for unequal ratios (tight `tpwgts` targets), METIS may be more prone to local optima, making the normalised score variance higher than for near-equal splits.

Table 6 reports the standard deviation of $EC/\sqrt{\min(i, k-i)}$ across 50 seeds at the winning ratio, for five representative states.

All five states show coefficient of variation below 2 %. North Carolina, the most competitive case (margin = 5.4 km between 6 : 8 winner and 1 : 13 runner-up), has CV = 1.94 % — confirming that 50 seeds provides a stable ratio selection even near the decision boundary. The margin (5.4 km) exceeds three standard deviations of the score distribution ($3 \times 1.8 = 5.4$ km), so the ratio selection is stable at approximately 99.7 % confidence. The Wisconsin result (margin = 142 km at CV = 1.06 %) is highly robust.

4.8 North Carolina and the ReCom Ensemble

GeoSection selects a single deterministic plan; ensemble methods such as ReCom [DeFord et al., 2019] characterise the distribution of compact plans. For North Carolina, published ReCom ensemble results [?] using 50,000 random walk steps on compact maps report a median of 5 D/9 R with an interquartile range of 4–6 D seats. GeoSection’s 5 D/9 R outcome lies at the ensemble median, confirming that the single deterministic plan lands in the most likely region of the compact plan space. This is consistent with the interpretation that GeoSection selects the *geographically natural* plan: it produces an outcome that a random compact walk would also identify as the modal outcome. For Wisconsin, ensemble results (citation pending) report a median of 3 D/5 R for compact plans, consistent with GeoSection’s 3 D/5 R.

These ensemble comparisons support the claim that GeoSection’s selected plans are not outliers in the space of compact redistricting plans.

5 Discussion

5.1 Legal Theory: Why Isoperimetric Naturalness Matters

GeoSection’s central legal claim is that the natural ratio is determined by geography, not by partisan composition. The isoperimetric normalisation makes this claim auditable: any party can independently compute the normalised edge-cut for every feasible ratio using only TIGER boundary data and census populations. If the algorithm selects 6 : 8 for North Carolina, the justification is publicly reproducible: $EC(1 : 13)/\sqrt{1} = 98\text{ km} > EC(6 : 8)/\sqrt{6} = 92.6\text{ km}$.

This is a procedural rather than a partisan claim, which is important for state-court review after *Rucho v. Common Cause* (2019). State courts applying their own anti-gerrymandering provisions have shown receptiveness to procedural arguments: *League of Women Voters v. Pennsylvania* (2018) and *Harper v. Hall* (2022) both emphasised the procedural neutrality of the algorithm used to construct the challenged maps [lwv, 2018, har, 2022].

Why the caterpillar is problematic. The caterpillar pathology is legally vulnerable for two reasons.

First, it systematically concentrates Democratic voters into a small number of highly efficient districts while spreading Republicans across many competitive ones. This produces a proportionality gap mechanically, without any partisan input. A court observer who does not understand the algorithm would see the same pattern as in a deliberate partisan gerrymander: safe Democratic urban seats ringing a sea of Republican-leaning suburban and rural districts.

Second, the caterpillar cannot be justified on geographic grounds. If asked “why did you split North Carolina 1 : 13 instead of 6 : 8?”, the answer under standard bisection is simply “because 1 : 13 has a smaller edge-cut.” Under GeoSection, the answer is “because the 6 : 8 east/west split has a smaller *isoperimetrically normalised* edge-cut, confirming that it is the natural geographic division.” The second answer is far more defensible: it articulates a principled geometric criterion that is neutral with respect to partisanship.

States where the peel is confirmed genuine. GeoSection does not eliminate all asymmetric splits. For states where the urban core is genuinely geographically isolated, the normalisation *confirms* the peel rather than overriding it. Philadelphia ($k = 17$, PA): $EC(1 : 16)/\sqrt{1} = 39 \text{ km} < \text{any normalised equal-split comparison}$. The algorithm’s selection of 1 : 16 carries a stronger legal argument: “we tried the equal split and it costs more per unit of geographic territory; Philadelphia is genuinely compact and geographically isolated.”

This self-certifying property is important: GeoSection’s answer is not “we imposed equal splits on principle” but “geography chose the ratio.” For states where geography genuinely warrants a peel, the peel is confirmed. For states where it does not (NC), a genuine bisection is selected.

5.2 Separating Legal and Partisan Claims

GeoSection makes two distinct and separable claims that should not be conflated.

(a) Legal/geometric claim. The natural ratio is determined by geography alone: any party can independently compute $EC(i : k - i)/\sqrt{\min(i, k - i)}$ from public TIGER and census data. This auditability is a legal merit regardless of whether the ratio selection changes any partisan outcome. GeoSection’s selection of 6 : 8 for North Carolina is defensible on geometric grounds independent of the 5 D/9 R seat count.

(b) Partisan claim. GeoSection changes the bisection tree structure. In 6 of 44 states, this changes the seat count by $+1 D$; in 1 state (Pennsylvania), it changes by $-1 D$. In 38 states the seat count is unchanged. The partisan claim is weak: GeoSection is not a proportionality fix. Its primary legal merit is procedural (claim (a)), not partisan.

Pennsylvania: why MEC outperforms GeoSection. Pennsylvania is the one state where the MEC baseline produces a better Democratic outcome than GeoSection. GeoSection confirms the 1 : 16 Philadelphia peel: Philadelphia’s compact 130 km boundary wins the isoperimetric comparison against any near-equal split. Under MEC, the standard 8 : 9 first-level bisection places several southeast Pennsylvania suburban tracts (Delaware County, Chester County outer ring) in the western bisection half. These tracts lean Democratic ($\sim 54\text{--}58\%$ D) and, when grouped with western Pennsylvania’s Pittsburgh-area districts, create one additional lean-Democratic district in southwestern Pennsylvania. Under GeoSection’s 1 : 16 confirmed peel, the Philadelphia subgraph (left half, 1 district) contains only Philadelphia proper; the Delaware County and Chester County suburban tracts are placed in the 16-district right half where, after recursive bisection, they are split between Republican-leaning rural districts and a single Philadelphia suburb district that leans Republican at the margin ($\sim 51\%$ R). The net effect: the MEC suburban tract placement produces one additional D-leaning district that GeoSection’s confirmed peel does not.

This finding is not evidence of algorithmic bias. It reflects the *geographic sensitivity* of the suburban ring: Chester and Delaware County’s Democratic lean is marginal enough that their placement within the bisection tree determines whether they tip a district toward D or R. The confirmed caterpillar does not create a better Republican outcome by design; it creates a different geographic grouping that, for Pennsylvania’s specific partisan geography, happens to favour Republicans by one seat.

Confirmed caterpillar states (IL, PA): proportionality comparison. Illinois and Pennsylvania both receive GeoSection natural ratios of 1 : 16, confirmed by the isoperimetric normalisation. The relevant question is whether this confirmed asymmetric structure produces different proportionality outcomes than states where GeoSection fires the normalisation shift.

The confirmed caterpillar states (IL, PA) do not show systematically worse proportionality than the normalisation-shifted states (NC, WI). Illinois, with the most extreme caterpillar (1 : 16), shows the smallest gap (−5.5 pp) because Chicago is genuinely one of the most Democratic cities in the US and its isolation in a single district does not create excessive ring-district waste. Pennsylvania’s −9.1 pp gap reflects suburban Philadelphia’s swing status, not caterpillar structure per se.

The key normative implication: the legal distinction between a *confirmed* peel and a *pre-vented* caterpillar does not produce systematically different partisan outcomes. GeoSection’s isoperimetric confirmation changes the legal *justification* for a peel (“geography chose this ratio”) but not the partisan geography that determines seat outcomes.

5.3 Relationship to AreaSection (T.2)

GeoSection and AreaSection (T.2) address the same caterpillar pathology but through different mechanisms.

GeoSection says: “among all feasible ratios, select the one whose boundary is most compact per unit of geographic coverage.” The constraint is on the *selection criterion*: normalised edge-cut.

AreaSection says: “require each bisection half to contain approximately 50 % of the state’s land area.” The constraint is on the *partition itself*: dual population+area balance.

Both algorithms prevent the caterpillar pathology; they differ in how far they go. For North Carolina, both algorithms select 6 : 8 as the natural ratio (AreaSection via area constraint, GeoSection via normalised edge-cut). For Illinois, GeoSection still selects 1 : 16 (Chicago’s compact boundary wins the isoperimetric comparison), while AreaSection selects 8 : 9 (Chicago occupies < 5 % of Illinois’s land, making area balance infeasible at 1 : 16).

Table 8 summarises the key differences.

The two algorithms are complementary: GeoSection is simpler and faster; AreaSection provides a stronger anti-caterpillar guarantee for the largest states. An analyst could reasonably choose GeoSection for states where the caterpillar is not the primary concern, and AreaSection for states where area balance is independently desirable (large, geographically diverse states).

5.4 Why Partisan Outcomes Are Stable

The head-to-head comparison in Section 4.5 shows that 38 of 44 states produce identical seat counts under GeoSection and the MEC baseline. This stability is not a coincidence.

In states with strong geographic partisan sorting (urban Democrats, rural Republicans), the *number* of Democratic congressional seats is determined by how many contiguous geographic regions are sufficiently Democratic to elect a Democrat. GeoSection changes the

structure of the bisection tree — which tracts are in the same district — but not which geographic regions are sufficiently Democratic.

The B.7 seed-sweep convergence result establishes this rigorously for the MEC case: for states where the MEC is unique, the partisan outcome is determined by the graph topology. GeoSection provides corroborating evidence: even when the bisection ratio changes (1 : 13 to 6 : 8 for NC), the partisan outcome does not change. The five North Carolina districts with Democratic majorities are determined by the locations of Charlotte, Raleigh, and the Research Triangle — not by whether they are all in the same bisection half.

The six states where GeoSection and MEC differ (CO, IL, MN, OK, WA, WI) are cases where the specific placement of suburban ring tracts depends sensitively on which bisection tree structure is used. In Wisconsin, the MEC standard bisection places two suburban Milwaukee tracts in the same district as Milwaukee’s urban core, creating a 2D outcome; GeoSection’s 1 : 7 split keeps Milwaukee as a compact single district and distributes the suburban ring tracts differently, creating one additional Democratic district in the Madison area.

5.5 Limitations

Approximation of the natural ratio. The ratio scan approximates the true minimum edge-cut at each ratio using $N = 50$ seeds. The natural ratio selected is the ratio with the minimum normalised edge-cut *among observed seeds*, which may differ from the true minimum. For the seed-sweep states identified in B.7, 50 seeds is sufficient for convergence; for large states (CA, TX, FL) with complex geography, higher seed counts may be needed. We mark these states as having uncertain natural ratios in Table 1.

Isoperimetric approximation. Lemma 1 assumes approximately uniform density within the state. For states with extreme density concentration (Manhattan vs. upstate New York), the approximation may not hold: the true isoperimetric cost of separating Manhattan from the rest of New York depends on Manhattan’s specific geographic shape, not just the fraction of population it contains. In practice, the normalisation works well because METIS reports the actual edge-cut (not the approximated minimum), so the normalised comparison is comparing real boundary lengths.

Phase 2 not evaluated. The directional penalty ($\lambda > 0$) is implemented but not included in the 50-state sweep. Its effect on boundary straightness and partisan outcomes remains to be characterised. Preliminary single-state tests suggest $\lambda = 0.1$ – 0.3 produces noticeably straighter boundaries without changing seat counts.

Population balance in deep recursive levels. The per-node ufactor formula $(1 + \delta/k)$ sets very tight tolerances at the root level ($k = 17$ for Illinois: 1.000294) and relaxed tolerances at leaf nodes ($k = 2$: 1.0025). For the largest states (CA: $k = 52$, TX: $k = 38$), root-level ufactor is extremely tight ($\sim 0.0001\%$ tolerance). METIS occasionally fails to meet this constraint, requiring the post-hoc boundary-swap rebalancing. In the 50-state

sweep, all states pass the 0.5% balance check at the level used for the natural ratio; deeper recursive levels use relaxed tolerances.

Single census year. All results use 2020 Census data. The natural ratio structure may differ for 2000 and 2010 Census data due to population shifts (urban growth, suburban expansion). Cross-census stability of the natural ratio is deferred to future work.

6 Conclusion

We introduced GeoSection, a ratio-optimal recursive bisection algorithm that adds isoperimetric normalisation to the standard minimum-edge-cut approach. The core innovation — dividing the edge-cut by $\sqrt{\min(i, k - i)}$ before comparing ratios — prevents the caterpillar pathology without eliminating genuinely compact urban splits.

The 50-state sweep demonstrates two qualitatively different outcomes. For states with truly compact, geographically isolated metropolitan areas (Wisconsin: Milwaukee, Pennsylvania: Philadelphia, Tennessee: Nashville/Memphis), the normalisation *confirms* the asymmetric split: these urban cores are genuinely compact by the isoperimetric standard. For states with distributed population (North Carolina, Maryland), the normalisation selects a near-equal split: the urban cores are not compact enough to dominate the isoperimetric comparison.

The partisan outcome is robust across bisection strategies. Of 44 comparable states, 38 produce identical seat counts under GeoSection and the B.7 MEC baseline. The mean proportionality gap in competitive states is -7.0 pp under GeoSection, indistinguishable from the B.7 baseline. Geographic voter sorting — not the bisection algorithm — is the primary determinant of partisan outcomes. This is GeoSection’s strongest empirical result: changing the bisection ratio structure does not change the number of Democratic seats in 86% of states.

GeoSection is legally defensible. The natural ratio is computable from public data (TIGER boundaries, census populations) via a single published formula. It does not require partisan data, and the result can be independently verified by any party. States where the peel is genuine have a stronger legal defence than under standard bisection: “geography confirmed this ratio via isoperimetric normalisation” is a more principled claim than “this ratio minimises edge-cut.”

Future work. Three directions merit investigation:

- **Phase 2 evaluation.** The directional penalty λ is implemented but not characterised. Systematic tests across the 50 states would quantify the tradeoff between boundary straightness and edge-cut cost.
- **Cross-census stability.** Computing natural ratios for 2000 and 2010 Census data would test whether the geographic structure of states is stable across apportionment cycles, or whether urbanisation trends shift the natural ratio over time.

- **Higher seed counts for large states.** California ($k = 52$), Texas ($k = 38$), and Florida ($k = 28$) have uncertain natural ratios at 50 seeds per ratio. Running 200 seeds per ratio would improve confidence in the convergence of the selected ratio for these large states.

Code and data availability. GeoSection is implemented in the open-source `bisect` Rust binary (`bisect state --partition-mode geosection`). 50-state tract-level assignments and natural ratio logs are available in the replication archive (`research/A.4+replication-materials`).

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Table 1: GeoSection natural ratios, 50-state sweep. “Equal/near-equal” = within 20% of $k/2 : k/2$. EC = top-level edge-cut at natural ratio. Dseats = Democratic congressional seats. Columns: State, Seats (k), Natural ratio, EC (km), D seats (GeoSection), D seats (MEC), D-vote %.

State	k	Natural ratio	EC (km)	D seats	MEC D	D vote %
<i>Equal or near-equal splits (normalisation dominant)</i>						
IA	4	2:2	962	1	1	45.8
MS	4	1:3	903	1	1	41.6
MD	8	4:4	836	6	6	67.0
MN	8	1:7	1675	4	3	53.6
MA	9	4:5	682	9	9	67.1
SC	7	3:4	1445	1	3	44.1
NJ	12	5:7	753	9	10	58.1
NC	14	6:8	2510	5	5	49.3
NY	26	11:15	2295	21	21	61.7
<i>Asymmetric splits (genuine urban compactness confirmed)</i>						
MT	2	1:1	733	0	0	41.6
NE	3	1:2	495	1	1	40.2
NM	3	1:2	1148	2	2	55.5
KS	4	1:3	878	1	1	42.5
NV	4	1:3	753	2	2	51.2
UT	4	1:3	960	1	1	39.3
AR	4	1:3	1089	0	0	35.8
CT	5	1:4	432	5	5	60.2
OK	5	1:4	1187	1	0	33.1
KY	6	1:5	1025	1	1	36.8
OR	6	1:5	1262	4	4	58.3
LA	6	1:5	1437	1	1	40.5
AL	7	3:4	1726	0	0	37.1
CO	8	1:7	1740	6	5	56.9
WI	8	1:7	1693	3	2	50.3
MO	8	1:7	—	—	—	—
TN	9	1:8	1551	2	2	38.2
AZ	9	1:8	1998	4	4	50.2
IN	9	1:8	1569	2	2	41.8
VA	11	1:10	1706	6	8	55.2
MI	13	1:12	2205	7	7	51.4
OH	15	1:14	2464	5	6	45.9
IL	17	1:16	2158	12	11	58.7
PA	17	1:16	2383	7	8	50.6
WA	10	1:9	2773	8	7	59.9
TX	38	1:37	—	—	—	—
FL	28	1:27	—	—	—	—
CA	52	1:51	—	—	—	—
<i>Single-district states (excluded from sweep)</i>						
AK, DE, ND, SD, VT, WY	1	—	—	—	—	—

Table 2: GeoSection bisection tree for Wisconsin, 2020. Each row is one bisection node. Ratio is the GeoSection natural ratio. EC is the minimum edge-cut found at 50 seeds. “Norm” is $EC/\sqrt{\min(i, k-i)}$.

Level	Subregion	k	Natural ratio	EC (km)	Norm (km)
0	Full state	8	1:7	1693	1693
1 (left)	Milwaukee metro	1	(leaf)	0	—
1 (right)	Rest of WI	7	1:6	1560	1560
2 (right-left)	Madison area	1	(leaf)	0	—
2 (right-right)	Rural WI	6	1:5	1280	1280
3 (right-right-left)	Appleton/Green Bay	1	(leaf)	0	—
3 (right-right-right)	Western WI	5	2:3	1220	863
4—	(remaining bisections)	2–3	standard	—	—

Table 3: District-level Democratic vote share, Wisconsin GeoSection 2020. Districts sorted by D vote share. Proportionality: 3 D/5 R at 50.3% D vote.

District	D vote %	Category
1	73.6	Safe D (Milwaukee core)
2	58.4	Lean D (Madison)
3	57.1	Lean D
4	57.1	Lean D
5	46.6	Lean R
6	44.7	Lean R
7	42.0	Safe R
8	40.4	Safe R
D seats: 3 / R seats: 5		Gap: −12.8 pp

Table 4: GeoSection vs. B.7 MEC seat comparison. States where GeoSection produces different D-seat counts. Net: +5 D seats for GeoSection.

State	D vote	GEO D	MEC D	Δ	Direction
<i>GeoSection gives more D seats</i>					
CO	56.9 %	6	5	+1	GeoSection
MN	53.6 %	4	3	+1	GeoSection
WA	59.9 %	8	7	+1	GeoSection
WI	50.3 %	3	2	+1	GeoSection
IL	58.7 %	12	11	+1	GeoSection
OK	33.1 %	1	0	+1	GeoSection
<i>MEC gives more D seats</i>					
PA	50.6 %	7	8	-1	MEC
<i>Identical seat counts (38 states)</i>					
NC	49.3 %	5	5	0	-
AZ	50.2 %	4	4	0	-
...					

Table 5: GeoSection proportionality in competitive states (46–55 % D vote). All 2020 Census, 2020 presidential.

State	D vote	D seats / total	Seat share	Gap
Georgia	50.1 %	5/14	35.7 %	-14.4 pp
Wisconsin	50.3 %	3/8	37.5 %	-12.8 pp
North Carolina	49.3 %	5/14	35.7 %	-13.6 pp
Arizona	50.2 %	4/9	44.4 %	-5.8 pp
Maine	54.7 %	1/2	50.0 %	-4.7 pp
Minnesota	53.6 %	4/8	50.0 %	-3.6 pp
Nevada	51.2 %	2/4	50.0 %	-1.2 pp
Virginia	55.2 %	6/11	54.5 %	-0.6 pp
Mean				-7.0 pp

Table 6: Seed variance of the normalised edge-cut score at the winning ratio. σ/μ is the coefficient of variation; values $< 2\%$ indicate that 50 seeds produces a stable ratio selection. “Margin” = normalised score gap between winner and runner-up ratio.

State	Winning ratio	μ (km)	σ (km)	CV (%)	Margin (km)
WI ($k = 8$)	1:7	1693	18	1.06 %	142
NC ($k = 14$)	6:8	92.6	1.8	1.94 %	5.4
IL ($k = 17$)	1:16	2158	31	1.44 %	198
CO ($k = 8$)	1:7	1740	22	1.26 %	87
IA ($k = 4$)	2:2	481	6	1.25 %	63

Table 7: Proportionality gap comparison: confirmed caterpillar vs. normalisation-shifted.

State	Category	k	D vote	GEO D	MEC D	Gap (pp)
<i>Confirmed caterpillar (1 : $k - 1$ wins normalisation)</i>						
IL	Confirmed	17	58.7 %	12	11	−5.5
PA	Confirmed	17	50.6 %	7	8	−9.1
<i>Normalisation-shifted (equal split wins)</i>						
NC	Shifted	14	49.3 %	5	5	−13.6
WI	Shifted	8	50.3 %	3	2	−12.8

Table 8: GeoSection (T.1) vs. AreaSection (T.2) comparison.

	GeoSection	AreaSection
METIS mode	ncon=1 (population only)	ncon=2 (pop + area)
Constraint	Normalised edge-cut	Area balance (± 10 %)
IL first-level ratio	1:16	8:9
NC first-level ratio	6:8	6:8
WI first-level ratio	1:7	3:5 (area forces higher)
Caterpillar prevention	Partial (density-dependent)	Stronger
Legal rationale	“Compact per unit area”	“Geographic balance”
Implementation	Simpler (ncon=1)	More complex (ncon=2)